

Internet Layer

The internet layer, sometimes called the network layer, is the second layer of the TCP/IP model responsible for logical addressing and ensuring the delivery of data to destination host. In this layer, IP addresses are attached to the packets to indicate the location of the sender and receiver.

The main protocol in this layer is the Internet Protocol

IP addresses

An internet protocol address, or IP address, is a unique string of characters that identifies the location of a device on the internet. Each device on the internet has a unique IP address. There are two types of IP addresses: IP version 4 (IPv4) and IP version 6 (IPv6). IPv4 addresses are made up of four decimal numbers separated by periods, each number ranging from 0 to 255. IPv6 addresses are made of eight hexadecimal numbers separated by colons, each number consisting of up to four hexadecimal digits. IP addresses can either be public or private. Private IP addresses are only accessible by other devices on the same local network. Public IP addresses are unique and routable on the internet. It is usually assigned through the Internet Service Provider.

Source IP is the IP address of the client's device on its local network. It tells the destination where to send the response back.

Destination IP is the IP address of the target secure web server resolved via DNS.

IP address vs MAC address

A MAC address is a unique alphanumeric identifier that is assigned to the network interface cards of devices on a network. This is also referred to as the physical address of the device. The first 24 bits are vendor specific and the last 24 bits are device specific.

	IP Address	MAC Address
Other Name	Logical address of a device	Physical address of a device
Layer	Operates in the Internet Layer (TCP/IP Layer 2)	Operates in the Network Access Layer (TCP/IP Layer 1)

Purpose	Identifies where the host is on the global network	Identifies the specific physical NIC on the local link
Example	192.168.1.50 (IPv4), 2002:db8::1 (IPv6)	A1 : B4 : C5 : DD : C1 : 3E

Journey of a data packet from the client's laptop to a web server.

1) Encapsulation on the user's laptop.

- When a user types in the url of the destination website, the web browser generates an HTTP/HTTPS request for the content on the web server – **Application Layer**
- The application data is broken into segments and a header containing the source and destination port numbers (Port 80 / 443) are attached to the segments – **Transport Layer**
- Headers containing the source and destination IP address are then attached to the segments making them packets – **Internet Layer**
- The packets are then prepared for transmission over the local physical medium. Frame headers containing the source and destination MAC addresses are added to the packets. The destination MAC address is the local network's router or default gateway – **Network Access Layer**

2) Transit Across the Internet

- The local router receives the frame, removes the frame header to access the destination IP address and then determines the next hop router using its routing table.
- The local router attaches a new frame header which contains the MAC address of the next hop router as the destination and sends it out to the Internet Service Provider.

3) Decapsulation at the Web Server

When the frame reaches the web server's network the initial encapsulation process occurs in reverse.

- The server's Network Interface Card (NIC) receives the frame, verifies that the destination MAC address matches its interface and then strips the frame header. The IP packet is then passed up to the next layer – **Network Access Layer**

- The server inspects the destination IP address to confirm that the packet is addressed to it, strips the IP header and passes the TCP segment up – **Internet Layer**
- The server checks the destination port, strips the TCP header and sends the data to the web server software – **Transport Layer**
- The web server software processes the request and prepares the requested webpage data then sends the response to the user's laptop using a reverse of the same path – **Application Layer**

Packet Journey Diagram



